ANNA Y. CHEN

Curriculum Vitae

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Legal Name: Yiyu Chen (陈一语)
Work Authorization: U.S. Permanent Resident
Fellowship Applications in Preparation: NSF GRFP; DOE CSGF; AMS Graduate Fellowship

Education

2023–2027 **B.A. in Climate System Science & B.A. in Applied Mathematics**, *Columbia College, Columbia University*, New York, USA.

GPA : 4.02/4.00

Research Experience

Lamont-Doherty Earth Observatory (LDEO), Columbia University

Spring 2026 – Present **[Senior Thesis] Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO_2 Reconstruction Skill in Climate Model Testbeds**.

Evaluated 5 Global Ocean Biogeochemical Models and 6 Earth System Models against independent pCO_2 data; assessed how choosing each as the testbed affects the skill of 7 stats- and ML-based data products.

Mentor : **Dr. Galen McKinley**, *Professor, Dept. of Earth and Environmental Sciences and LDEO*, Columbia U. (Lab web-page)

Spring 2025 – Present **Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure**.

Developed gap-filling data product and animation for global sea-surface temperature (SST) using sparse deep-ocean core data over the Last Interglacial; coding in Python. NSF-funded project: HISEAS

Mentor : **Dr. Jerry McManus**, *Professor, Dept. of Earth and Environmental Sciences; Deputy Director for Education, LDEO, Earth and Environmental Sciences*, Columbia U.

Summer 2024 – Spring 2025 **Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity**.

Visualized and analyzed seasonal and geospatial patterns of satellite-derived marine net primary production based on sea-surface temperature and pCO_2; coding in R.

Practiced phytoplankton culturing, flow-cytometry, media making, and sterile lab methods.

Mentor : **Dr. Sarah Hurley**, *Assistant Research Professor, LDEO*, Columbia U. (Lab web-page)

Manuscripts in Preparation

Chen, A. Y., McKinley, G. A., Heimdal, T. H., Fay, A. R. *(2026)*. Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO_2 Reconstruction Skill in Climate Model Testbeds. *In progress.*

Heimdal, T. H., Ford, D. J., Keppens, M., Rogerson, J., Barton, A., Chevallier, F., Roobaert, A., Fay, A. R., McKinley, G. A., Gehlen, M., **Chen, A. Y.**, Shutler, J. D., & Landschützer, P. *(2026)*. The SOCOMv2 initiative: comparison of surface ocean pCO_2 data products using the CMIP6-testbed.

Manuscript in preparation for submission.

Holm, H. C., Schaum, A. P., Ellis, A., Acosta, B., **Chen, A. Y.**, Wing, B., McKinley, G. A., Dyhrman, S. T., & Hurley, S. J. *(2026).* Using Climate Models to Inform Multi-Driver Experiments with Future Ocean Conditions: Applications to Phytoplankton. *Manuscript in preparation for submission to Global Change Biology.*

Fellowships & Awards

Summer 2026 **Comer Climate Undergraduate Summer Research Fellowship**, *Columbia U.*, ($9,600).
Funds 4 Columbia College students to participate in a 10-week supervised summer research related to climate, sustainability, and environmental justice.

Summer 2025 **Columbia College Summer Research Fellowship**, *Heinrich Family Global Fellowship*, ($3,600).
Awarded to non-seniors at Columbia College for academic excellence to pursue 6-weeks of independent summer research with faculty mentorship

Spring 2025 **Data Science Institute Scholars Program**, *Columbia U.*, ($3,000).
Awarded to mostly Master's, and some highly achieving Bachelor's students, to pursue mentored research in a data-science-related field during an academic semester (15 weeks)

2024-2025 **Laidlaw Undergrad. Research and Leadership Scholarship**, *Columbia U.*, ($7,600 + housing).
Funds Columbia College first-years who wish to pursue summer research, community engagement, and leadership across all disciplines. Full-time stipend awarded across two summers, 6-weeks in each

All Semesters **Dean's List**, *Columbia U.*
Awarded to students with GPA 3.6 and above

2023 **High School Valedictorian**, *YK Pao School*, Shanghai, China.
Received $28,000 in merit-based scholarship. Voted as Class of 2023 graduation ceremony speaker.

Teaching Assistantship

Spring 2027 **EESC GR4926: Intro to Chemical Oceanography**, Columbia U.

Fall 2025 **EESC UN2100: Earth's Environmental Systems: Climate**, Columbia U.

Presentations

Fall 2026 **AGU 2026 Annual Meeting**, San Francisco.
1. Oral Presentation: "Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO_2 Reconstruction Skill in Climate Model Testbeds" *(Session **A24F**)*
2. Poster: "Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure" *(Session **PP21C**)*

Fall 2026 **Undergraduate Research Symposium**, Columbia U.
Poster: Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO_2 Reconstruction Skill in Climate Model Testbeds

Summer 2026 **McKinley/Lovenduski Group Meeting**, U. of Colorado-Boulder.
Talk: "Evaluating the Impacts of Testbed Model Choice to Global Ocean Surface pCO_2 Reconstruction"

Spring 2026 **Mini-Symposium on AI-based Science**, LDEO.
Talk: "Can Selecting Testbed (CMIP6) Models Based on the Representation of Seasonality Improve Global Surface-Ocean pCO_2 Modeling?"

Fall 2025 **Geochemistry Poster Session**, LDEO.
Poster: "Last Interglacial Sea-Surface Temperature Reconstruction: A Preliminary Approach Using Reduced-Space Optimal Interpolation"

Fall 2025 **North America Laidlaw Scholars Conference**, Brown U.
Talk: "Farming Futures: Building Colorado Communities Through Environmental Stewardship"

Spring 2025 **Scholars Program Poster Presentation**, *Data Science Institute*, Columbia U.
Poster: "Visualizing Sparse Sea Surface Temperature Data From the Last Interglacial"

Fall 2024 **North America Laidlaw Scholars Conference**, Columbia U.
Talk: "Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity"

Fall 2024 **LDEO 75th Anniversary Mini Symposium**, LDEO.
Fall 2024 **Undergraduate Computer and Data Science Research Fair**, *Data Science Institute*, Columbia U.
Fall 2024 **Undergraduate Research Symposium**, Columbia U.
Poster for all 3 above: "Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity"

Leadership and Volunteering

Spring 2024 – Present **President**, *Zen Buddhism Club*, Columbia U..
Organize meditation retreats weekly at Columbia, quarter-annually at Columbia, NYU, & Princeton (≈120 ppl.)

Fall 2024 – Present **President**, *Marie Tharp Society*, Dept. of Earth and Environmental Sciences, Columbia U.
Host social mixers, faculty Q&As, geology movie nights, and student-led Lamont Tours

Fall 2025 – Present **Vice President**, *Columbia U. Cycling Team.*
Coordinate training and racing, apply for school recognition, recruit members, oversee WhatsApp group chat

Fall 2025 – Present **Volunteer**, *Climate Lab*, American Museum of Natural History.
Explain climate processes to museum visitors through hands-on experiments, e.g. the thermohaline circulation

Fall 2024 & 2026 **Volunteer**, *Open House*, LDEO.
Educated children on paleo-botany through making bookmarks from plant-extract paper chromatography

Summer 2024 **Volunteer**, *World Oceans Day*, United Nations.
Facilitated guest check-in and backstage set-up for event with 30+ speakers and 350+ attendees at UN Headquarters

Languages and Skills

Natural Chinese (native), English (fluent), German (elementary)
Computational Python, R, LaTeX, Markdown
Software Google Suite, MS 365 (Microsoft Suite), iWork (Apple Suite)